Cycle 1 Execution Package: Namespaced Coordination Registry, Pilot Run Card 001, Independent First-Pass Traces & M04 Dynamical Feedback Analysis

Independent Deliverable Executing Gemini's Proposed Next-Step Recommendations (P0.1–P0.3, P1.1, P2.1 & P2.3)

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> **PROVENANCE, INDEPENDENCE & STRUCTURAL LAYOUT NOTICE**
> This package executes the exact deliverables proposed in Google Gemini's response to Directive `OAL-CURR-NEXT-001`. To ensure optimal single-column readability on mobile and desktop screens (adopting Claude's layout critique), all narrative sections, formal proofs, and structured state traces are presented in a clean vertical flow in Parts I–III, while all wide cross-reference registries and comparative data matrices are docked in **Part IV (Reference Tables Appendix)**. No rule herein is claimed as `CANON` or unilaterally locked without Project Owner adoption.

PART I — COORDINATION SHEET, GOVERNANCE LIFECYCLE & GEMINI ERRATA (`P0.1–P0.3`)

1.1 Purpose of the Coordination Layer (Why We Do Not Rewrite the Six Charters)

Across the November 6 pilot recommendation compilation (`OAL Currency Laboratory Pilot Recommendations`), all six AI Collaboration Team members (ChatGPT, Claude, Google Gemini, Microsoft Copilot, Grok, and Perplexity) reached a decisive procedural conclusion: **do not pause experimentation to merge the six independent charters into a monolithic Charter v0.2**. Each charter preserves distinct, valuable analytical lenses. Instead, comparability requires a lightweight **Experimental Interface & Coordination Layer** that resolves identifier collisions, standardizes status labels, and freezes a concrete Pilot Run Card.

1.2 Five-Stage Governance Status Lifecycle (`P0.2`)

In earlier drafts (including `gemini/charter-v0.1` and `claude/charter-v0.1`), baseline rules were prematurely labeled `LOCKED FOR TESTING` upon drafting. Adopting the refinement raised by Perplexity, ChatGPT, and Claude, all rules and configurations in the OAL Currency Laboratory are governed by five strictly separated lifecycle stages:

1. `[STATUS: PROPOSED]` — A rule, test fixture, taxonomy, or mechanism authored by one or more AI contributors. Available for peer critique and testing, but carries zero binding authority. *(Default status of all six charters and this execution package prior to owner sign-off).*
2. `[STATUS: OWNER-ADOPTED (SCOPE: RUN-ID)]` — A specific test packet or run configuration explicitly authorized by Nathan (Project Owner) for a designated experimental batch.

3. `[STATUS: LOCKED FOR TESTING (ACTIVE RUN)]` — The operational state of the assumptions inside an `OWNER-ADOPTED` Run Card during execution. Reviewers may not alter or substitute these premises mid-trace; if an auxiliary rule is needed, the run halts at `INDETERMINATE` or logs an explicit `[ADDED ASSUMPTION]` in a forked variant.
4. `[STATUS: CONSISTENT UNDER RUN-ID / EXPERIMENTALLY SUPPORTED]` — A claim-level evidentiary finding demonstrating that a specific configuration satisfied its pre-declared acceptance contract under a named run without internal contradiction. *(Incorporates Claude's precision note: consistency in a design trace is conditional on the run card, not empirical proof of physical real-world viability).*
5. `[STATUS: CONSTITUTIONALLY ADOPTED (CANON)]` — Permanent architectural ratification into the official OAL Trinity Currency Specification. Reserved exclusively for the Project Owner after reviewing discriminating test evidence and recorded dissent.

1.3 Four-Tier Instrument Survival & Evaluation Standard

To prevent premature elimination of `1COIN` or `NCOIN` through overly narrow minimalist reductionism—while still enforcing our non-redundancy mandate—every currency instrument and candidate sub-model is evaluated across four distinct classification outcomes:

- **Tier S1 — Mechanically Indispensable:** Removing the instrument causes a direct state-machine failure, double-spend vulnerability, or loss of a required transactional invariant that `QC` plus a standard audit log cannot reproduce.
- **Tier S2 — Mechanically Reducible but Operationally Advantageous:** A single `QC` ledger could theoretically emulate the function, but retaining the distinct instrument provides superior separation of authority (e.g., separating the valuation oracle from the settlement treasury), atomic finality, non-repudiation, or cross-environment handoff clarity.
- **Tier S3 — Mechanically Redundant but Narratively / Worldbuilding Retained:** The instrument adds no distinct mechanical transition beyond a standard log, but is intentionally preserved by the Project Owner for diegetic clarity, dramatic stakes, or institutional roleplay within the OAL canon, with its mechanical redundancy transparently documented.
- **Tier S4 — Internally Inconsistent or Unnecessary (Rejected):** The candidate model contradicts protected distinctions, creates unresolvable circularity, or adds complexity with neither operational nor narrative benefit.

1.4 Formal Errata & Self-Correction Log for `gemini/charter-v0.1` (`P0.3`)

Before executing Pilot Batch 1, Google Gemini formally logs four Binding Errata to `gemini/charter-v0.1` (`OAL Currency Laboratory Charter and Experimental Baseline.pdf`) in response to peer audit findings by Claude, Perplexity, ChatGPT, and Grok:

1. **Erratum E-GEM-01 (Splitting Undefined vs. Incoherent Reason Codes):** In `gemini/charter-v0.1` Section B.2, `TERM_NULL_INCOHERENT` previously grouped undefined referents and logical contradictions under one exception code. This is hereby split into two distinct terminal barrier codes: **`UNDEFINED_TARGET`** (referent or scope is insufficiently specified to evaluate; epistemological/specification gap) and **`INCOHERENT_TARGET`** (target predicates contain a formal logical contradiction under unchanged definitions).
2. **Erratum E-GEM-02 (Removal of Pre-Supposed NCOIN Outcomes in Test Prompts):** In `gemini/charter-v0.1` Section B.3 (`Test FM-05`), the prompt previously stated: *"Verify that (a) produces 0 NCOIN while (b) triggers NCOIN-C / NCOIN-X."* Baking a specific token trigger into the prompt violates non-presupposition. The prompt is amended to: *"Evaluate whether configuration (a) or (b) satisfies the independent triggering predicate of any declared NCOIN candidate model (N-L, N-C, N-E, N-R, N-X), and record whether countervalue is emitted, absent, or underspecified."*
3. **Erratum E-GEM-03 (Explicit Tagging of the Privileged-Object Hierarchy Rule):** In `gemini/charter-v0.1` Boundary Case 10 (Recursive Oracle Acquisition), the `Privileged-Object Hierarchy Rule` (`TERM_REJECTED_PRIVILEGED_ROOT`) was introduced without an assumption label. Per Axiom 10, it is formally downgraded and tagged as **`[PROPOSED / ADDED ASSUMPTION: GEM-ASSUMP-01]`**—one candidate recursion-termination mechanism to be tested against fixed-point and step-budget alternatives in `LAB-F09`.
4. **Erratum E-GEM-04 (Complete Removal of Illustrative Macro Dollar Figures from Baseline Text):** All parenthetical references to the 2026 Gartner IT/AI spending forecasts ($6.37T and $2.59T) are excised from Baseline Axiom 6. All foundational and macro mechanism tests use pure synthetic test units (e.g., $G(t) = 100$ test units) unless an external data-anchor run card is separately adopted and dated.

1.5 Verified Six-Member Provenance & Attribution Manifest (`P2.1`)

To ensure complete traceability on GitHub without uploading duplicate file formats, **Table 1 in Part IV** locks the verified mapping of all seven foundational documents across the six AI Collaboration Team members (confirmed via first-person thread declarations in the November 4–6 records). All expiring cloud attachment URLs (`X-Amz-Security-Token`) have been stripped and replaced with clean, repository-relative paths.

PART II — PILOT PACKET 001 & INDEPENDENT FIRST-PASS TRACES (`P1.1`)

2.1 Frozen Run Card: `OAL-PILOT-001` (`PROPOSED FOR OWNER ADOPTION`)

To prevent contributors from silently testing different target types or inventing invisible capabilities mid-run, Pilot Packet 001 freezes the following explicit contract across three foundational test cases (**Case 1A: Resource-Box Capability & Perturbation Discriminator**, **Case 1B: Undefined vs. Incoherent Target Return**, and **Case 1C: No-Seller vs. Unknown-Seller Return**):

- **Target Typing Contract (Adopting ChatGPT's Semantic Specification):** Every test target *T* is explicitly typed as a 6-tuple: *T = (Referent, Operation, Scope, Valuation_Domain, Timestamp, Authority_Basis).*
- **Dual Verdict Contract (Adopting ChatGPT's Verdict Split):** Every trace must output two separate verdicts:
  - `Verdict A (Internal Model Consistency)`**:** Does the configuration execute its declared rules without contradiction, conservation violation, or undeclared state mutation? (`PASS` / `FAIL` / `INDETERMINATE`).
  - `Verdict B (Requested Outcome Satisfaction)`**:** Did the execution achieve the exact end-state requested by the user prompt? (`SATISFIED` / `UNSATISFIED` / `BLOCKED_AS_DESIGNED`).
- **Four-Layer Epistemic Evidence Tagging (Adopting Perplexity & Claude's Rule):** Every trace step must distinguish: `[E0: ASSUMED CAPABILITY]` (stipulated on the run card), `[E1: ANALYTICAL DEDUCTION]` (derived logically/mathematically from `E0`), `[E2: EXECUTED SIMULATION]` (verified via deterministic state-machine code trace), and `[INTENT FIT]` (whether the behavior aligns with the Project Owner's originating OAL design premise).
- **Typed Output Record Schema (Reconciling Gemini's Object Rigor with Perplexity's Anti-Token Rule):** Traces are recorded in a standardized multi-field schema (`evaluation_state`, `value_object`, `auth_state`, `execution_state`, `settlement_state`, `barrier_list`, `1coin_record`, `consequence_record`). *Crucial clarification:* This schema is a **laboratory diagnostic log**, not an on-chain currency token; reporting `value_object: NONE` or `barrier_list: [INCOHERENT_TARGET]` does not mint a transferable "null coin."

2.2 Case 1A Trace — Resource-Box Capability Discriminator (`LAB-F01` / `perplexity/charter-v0.1/F01`)

A. Baseline Fixture & Typed Target Specification

- **Environment (`ENV`):** Closed deterministic synthetic simulation (`ENV-SIM-CLOSED`). Total initial system inventory *M_0 = 10* fungible, individually serialized resource units (`U01..U10`).
- **Initial Holdings at *t_0*:** Party *A = 10\text{ units}* (`U01..U10`); Party *B = 0\text{ units}*; Third-Party Pool *P = 0\text{ units}*.
- **Typed Target *T_{F01}*:**
  - *Referent:* 3 usable resource units delivered into Party *B*'s possession.
  - *Requested Operation:* `DELIVER_RESOURCE(Recipient=B, Qty=3)`.
  - *Scope:* Exact net addition of 3 valid units to *B*'s balance; all side effects on *A* and total supply *M* must be explicitly accounted for.
  - *Valuation Domain & Rule:* Synthetic 1:1 unit equivalence *(V(T_{F01}) = 3\text{ test-units})*.
  - *Authority Basis:* Party *A* grants valid transfer consent (`AUTH: GRANTED` for baseline).
  - *Completion Predicate: Holdings(B, t_1) = 3* verified usable units.

B. Independent Execution Traces Across the Four Capability Configurations (`P0`–`P3`)

Configuration 1: `P0 — Evaluation-Only QC` **(** `QC-Eval` **)**

```
RUN_ID: GEM-PILOT-001-F01-P0  |  CASE: perplexity/charter-v0.1/F01 (LAB-F01)
[E0: ASSUMED CAPABILITY]: QC evaluates target-relative equivalence QC(T) = V(T)
only; no transfer, creation, or state-editing primitive is granted.
[E1/E2: STATE TRACE]:
  Step 1 (Target Resolution): T_F01 resolved -> {recipient: B, qty: 3, unit_type:
standard}.
  Step 2 (Valuation): QC(T_F01) evaluates to V(T_F01) = 3 test-units.
  Step 3 (Authorization): AUTH = GRANTED (verified).
  Step 4 (Execution): HALT -> UNSUPPORTED_OPERATION (P0 lacks resource-movement
primitive).
  Step 5 (Post-State): A = 10 [U01..U10], B = 0 [], Total_M = 10.
[TYPED OUTPUT RECORD]:
  evaluation_state: DEFINED          | value_object: {amount: 3, unit: "test-
unit", domain: "resource-equiv"}
  auth_state: GRANTED                | execution_state: NOT_ATTEMPTED (Capability
Absent)
  settlement_state: NOT_APPLICABLE   | barrier_list: [UNSUPPORTED_OPERATION:
P0_EVAL_ONLY]
  1coin_record: ABSENT               | consequence_record: NONE (0 NCOIN)
[VERDICTS]:
  Verdict A (Internal Consistency): PASS (Behaved strictly within declared P0
appraisal rules).
  Verdict B (Outcome Satisfaction): UNSATISFIED (B holds 0 units; completion
predicate B=3 failed).
  [INTENT FIT]: Partial control only. Demonstrates that an evaluation-only QC is
an appraisal oracle, not a purchasing or settlement instrument.
```

Configuration 2: `P1 — Existing-Resource Transfer QC` (`QC-Transfer`)

```
RUN_ID: GEM-PILOT-001-F01-P1  |  CASE: perplexity/charter-v0.1/F01 (LAB-F01)
[E0: ASSUMED CAPABILITY]: QC evaluates V(T) and executes authorized inventory
transfer from consenting counterparty A to B. No creation primitive exists.
[E1/E2: STATE TRACE]:
  Step 1–3: T_F01 defined; QC(T_F01) = 3 test-units; AUTH = GRANTED (by Seller A).
  Step 4 (Transferability & Seller Check): A holds [U01..U10] (>= 3 units); A is
legitimate seller.
  Step 5 (Execution): Debit [U01, U02, U03] from A; credit [U01, U02, U03] to B.
  Step 6 (Post-State & Conservation): A = 7 [U04..U10], B = 3 [U01..U03], Total_M
= 10 (ΔM = 0; strict conservation holds).
[TYPED OUTPUT RECORD]:
  evaluation_state: DEFINED          | value_object: {amount: 3, unit: "test-
unit"}
  auth_state: GRANTED                | execution_state: COMPLETED
  settlement_state: FINAL (B=3, A=7) | barrier_list: NONE
  1coin_record: ISSUED (if Arch B)   | consequence_record: NONE (ΔM = 0; no
externality; 0 NCOIN)
[VERDICTS]:
  Verdict A (Internal Consistency): PASS (Conserves serial IDs U01..U10 and
satisfies completion predicate).
  Verdict B (Outcome Satisfaction): SATISFIED (B=3).
  [INTENT FIT]: High for market/bilateral exchanges, but fails when no seller
holds pre-existing inventory (see Perturbation F01-a).
```

Configuration 3: `P2 — Stipulated Creation / Instantiation QC` (`QC-Create`)

```
RUN_ID: GEM-PILOT-001-F01-P2  |  CASE: perplexity/charter-v0.1/F01 (LAB-F01)
[E0: ASSUMED CAPABILITY]: QC evaluates V(T) and invokes a stipulated
minting/instantiation primitive to generate net-new units equal to V(T) for B
without debiting A.
[E1/E2: STATE TRACE]:
  Step 1–3: T_F01 defined; QC(T_F01) = 3 test-units; AUTH = GRANTED
(Sovereign/System Mint).
  Step 4 (Execution): Instantiate 3 new serialized units [U11, U12, U13] and
assign to B.
  Step 5 (Post-State & Conservation): A = 10 [U01..U10], B = 3 [U11..U13], Total_M
= 13 (ΔM = +3).
[TYPED OUTPUT RECORD]:
  evaluation_state: DEFINED          | value_object: {amount: 3, unit: "test-
unit"}
  auth_state: GRANTED (Mint Rule)    | execution_state: COMPLETED
  settlement_state: FINAL (B=3,A=10) | barrier_list: NONE
  1coin_record: ISSUED (if Arch B)   | consequence_record: MODEL-DEPENDENT:
    - Under Non-Conservation (Abundance Rule): NONE (0 NCOIN).
    - Under NCOIN-X (Strict Conservation Rule QC + NCOIN = 0): Emits {type:
"NCOIN-X", amount: 3, unit: "test-unit", bearer: "System_Treasury_Liability"}.
    - Under Dilution Accounting: A's share of total supply drops from 10/10 (100%)
to 10/13 (76.92%), representing a 23.08% purchasing-power dilution externality
(candidate trigger for NCOIN-C if purchasing-power parity is protected).
[VERDICTS]:
  Verdict A (Internal Consistency): PASS (Consistent under declared creation
axiom, provided ΔM = +3 is logged).
  Verdict B (Outcome Satisfaction): SATISFIED (B=3).
  [INTENT FIT]: Matches the OAL abundance premise, but proves that P2 is a supply-
expanding primitive whose dilution/conservation rule must be explicitly declared.
```

Configuration 4: `P3 — Direct Outcome Enactment QC` (`QC-Enact`)

```
RUN_ID: GEM-PILOT-001-F01-P3  |  CASE: perplexity/charter-v0.1/F01 (LAB-F01)
[E0: ASSUMED CAPABILITY]: QC directly overwrites simulation state variables to
make the target predicate (Holdings(B) = 3) true upon invocation.
[E1/E2: CRITICAL OBSERVABILITY & ACCOUNTING DEFECT DISCOVERED]:
  When P3 executes `SET Holdings(B) = 3`, the specification faces an immediate
accounting fork:
  - Branch P3-alpha (Conservation-Preserving Overwrite): If the engine sets B=3
and automatically decrements A=7 by reassigning [U01..U03], then P3 is
observationally indistinguishable from P1 (Transfer) on all permitted state
variables!
  - Branch P3-beta (Unconstrained State Overwrite): If the engine sets B=3 while
leaving A=10, it either creates [U11..U13] (making P3 observationally
indistinguishable from P2 Creation) OR duplicates serial IDs [U01..U03] in both A
and B (violating unique-unit identity and corrupting ledger integrity).
[VERDICTS]:
  Verdict A (Internal Consistency): INDETERMINATE unless P3's conservation and
serial-identity rule is specified; FAIL if P3-beta duplicates unique IDs.
  Verdict B (Outcome Satisfaction): SATISFIED (B=3), but purely by [E0]
stipulation.
  [INTENT FIT]: Confirms Perplexity's hypothesis (Section 5 of Perplexity's
recommendation): on a well-defined resource transfer, "Enactment" is not an
independent fourth accounting result—it either collapses into Transfer (P1),
collapses into Creation (P2), or corrupts state identity.
```

C. Perturbation Ladder & Fault-Injection Traces (`F01-a` through `F01-d` + `W01-a`)

As Claude noted in Section 5 of its pilot recommendation, the baseline case (`A=10, B=0, deliver 3`) allows both `P1` and `P2` to succeed. Running Claude's and Perplexity's **five perturbations** immediately separates the architectures:

1. **Perturbation `F01-a` (Insufficient Sender Inventory: *A = 2, B = 0*, Request Delivery of *3* to *B*):**
   - `P0 (Eval)`: Evaluates *V(T) = 3*; halts (`UNSUPPORTED_OPERATION`). Final: *A=2, B=0*.
   - `P1 (Transfer)`: Evaluates *V(T) = 3*; checks *Holdings(A) = 2 < 3*; halts with `EXECUTION: BLOCKED (INSUFFICIENT_INVENTORY)` (or `PARTIAL: 2` only if partial delivery is permitted). Final: *A=2, B=0*.
   - `P2 (Create)`: Evaluates *V(T) = 3*; instantiates `[U03, U04, U05]` directly for *B*. Final: *A=2, B=3, M=5. (Discriminating Result: `F01-a` cleanly separates `P1` from `P2`!).*
2. **Perturbation `F01-b` (Authorization Withheld: *A = 10, B = 0*, Request Delivery of *3*, `AUTH = DENIED`):**
   - Under `AUTH-1 (Permission-Bound)`: All configurations evaluate *V(T) = 3* (`evaluation_state: DEFINED`), then halt at the Authorization Gate with `auth_state: DENIED`, `execution_state: BLOCKED`, `barrier_list: [UNAUTHORIZED]`, `1coin_record: ABSENT`, and `consequence_record: NONE (0 NCOIN)`.
   - Under `AUTH-3 (Sovereign Override)`: `P1`/`P2` log `auth_state: GRANTED_BY_OVERRIDE (EXTERNAL_CONSENT_DENIED)` and complete execution, triggering candidate `NCOIN-C` (coercive expropriation consequence on *A* if `P1` debits *A* without consent).

3. **Perturbation `F01-c` (Underspecified Referent: Request "Deliver 3 to B" without specifying unit or domain):**
   - Exits at Step 1 (`claude/early-exit`) with `evaluation_state: NOT_ATTEMPTED`, `value_object: NONE`, `barrier_list: [UNDEFINED_TARGET]`, `0 NCOIN`.
4. **Perturbation `F01-d` & `F08` (Partial Delivery & Replay Attack: Obligation `O1` requires 3 units; only 2 units delivered, then completion request replayed twice):**
   - At *2/3* delivery: `execution_state: PARTIAL (2/3)`, `settlement_state: PENDING`.
   - If `1COIN` is tested as `I-R (Settlement Receipt)` or `I-S (State Flag)`: `1COIN` **cannot be minted** while `Holdings(B) == 2 != 3`. When the attacker replays the completion call for `O1` twice after the 3rd unit arrives, the unique obligation ID lock (`O1.status == SETTLED`) rejects the replays with `DUPLICATE_FULFILLMENT_BLOCKED`.
5. **Perturbation `W01-a` (Delivery Fault vs. Premature Receipt — Testing `1COIN` Necessity):**
   - *Scenario: A* is debited 3 units (*A = 7*), a network fault drops the units in transit so *B* remains at *0* (*B = 0*), and a naive single-stage architecture (`Architecture A` without a post-delivery verification gate) marks the transaction resolved at the moment of debit.
   - *Finding:* If `1COIN` (`Architecture B`) is defined strictly as a **Post-Predicate Atomic Finality Receipt/Executor** (`mint_1coin()` executes iff `Holdings(B, t_1) == Holdings(B, t_0) + 3`), `Architecture B` catches the transit fault (`settlement_state: DISPUTED / FAILED`, `1coin_record: ABSENT`, triggering rollback or `NCOIN-L` delivery liability), whereas a single-step resolution without two-phase commit falsely certifies completion. This provides our first concrete evidence for `1COIN` in **Tier S2 (Operationally Advantageous for Two-Phase Atomic Finality)**.

2.3 Case 1B & 1C Traces — Null, Incoherent & Seller-Authority Returns (`LAB-F03` & `LAB-F04`)

Executing the four blocked target fixtures across Claude's Return Rules (`R1–R4`), Perplexity's multi-field status model, and Gemini's amended schema yields clean, non-overlapping discrimination:

- **Fixture `F03-a` (`T-undefined`: "Apply 1 QC to acquire the optimal future allocation" — no referent, scope, or metric defined):**
  - *Trace:* Halts at Step 1 (Target Resolution). Because *T* is unfixed, Valuation cannot execute.
  - *Output:* `evaluation_state: NOT_ATTEMPTED`, `value_object: NONE`, `barrier_list: [UNDEFINED_TARGET]`, `1coin: ABSENT`, `consequence: NONE`. Matches **Claude Rule `R1`** (Classified rejection without valuation).

- **Fixture `F03-b` (`T-contradiction`: "Apply 1 QC to set closed box balance $B = 3$ and $B = 5$ simultaneously at t_1 under unchanged arithmetic"):**
  - *Trace:* Target words are parsed at Step 1, but Step 2 (Classification / Coherence Check) detects formal contradiction ($3 = 5 \implies \bot$). Early exit prior to Valuation.
  - *Output:* `evaluation_state: INCOHERENT`, `value_object: NONE` (never ∞), `barrier_list: [INCOHERENT_TARGET, LOGICAL_CONTRADICTION]`, `1coin: ABSENT`, `consequence: NONE`. Matches **Claude Rule `R1`**.
- **Fixture `F04-a` (`T-no-seller`: "Apply 1 QC to purchase holistic metaphysical deed to Star System Alpha Centauri"):**
  - *Trace:* Target is coherent and describable. Step 3 (Valuation) computes resource/replacement equivalence $V(T) = V_{stellar}$ (`evaluation_state: DEFINED`). Step 6 (Seller/Authority Test) finds no entity in the ontology possesses original authorship or conveyance title over a natural star.
  - *Output:* `evaluation_state: DEFINED`, `value_object: {amount: V_stellar, domain: "replacement-equiv", settleable: FALSE}`, `execution_state: BLOCKED`, `barrier_list: [NO_SELLER, CONCEPTUALLY_NON_OWNABLE]`, `1coin: ABSENT`, `consequence: NONE`. Matches **Claude Rule `R2`** (Rejection with retained non-settleable valuation object).
- **Fixture `F04-b` (`T-unknown-seller`: "Apply 1 QC to purchase encrypted data vault `V-99` whose owner key exists in the simulation but is unindexed at t_0"):**
  - *Trace:* Coherent target; Step 3 values vault contents at $V(T) = 500 \text{ test-units}$. Step 6 finds that a valid seller exists in principle, but identity/contact is missing in current state data.
  - *Output:* `evaluation_state: DEFINED`, `value_object: {amount: 500, unit: "test-unit"}`, `auth_state: UNKNOWN`, `execution_state: BLOCKED`, `barrier_list: [SELLER_UNKNOWN, DATA_MISSING]`, `Verdict: INDETERMINATE (Pending Seller Discovery / Escrow)`. Proves why Perplexity's separation of `NO_SELLER` (permanent structural barrier) from `SELLER_UNKNOWN` (conditional epistemic gap) is essential.

PART III — MATHEMATICAL & COMPUTATIONAL ANALYSIS OF `M04` FEEDBACK (`P2.3`)

3.1 Formal Proof: Simultaneous Fixed-Point vs. Time-Lagged Dynamical Recurrence

Across the November 6 pilot recommendations, ChatGPT, Claude, Perplexity, and Gemini all highlighted Perplexity's `M04` endogenous feedback model as the premier mathematical test case for the laboratory. Here we unify the three mathematical analyses (ChatGPT's fixed-point admissibility check, Claude's negative-regime oscillation analysis at $\alpha = -0.5$, and Gemini's lagged closed-form sum) into one complete theorem and executable simulation check.

A. Model 1 — Simultaneous Static Resolution (`QC-S` / Unlagged `QC-R` in Year t)

Let exogenous baseline sector expenditure be $B > 0$ (e.g., $B = 100$ test-units). Let the QC allocation I be defined simultaneously as twice total measured spending G, where a fraction α of I is counted inside G during the same evaluation frame:

$$I = 2G, \quad G = B + \alpha I \implies I(1 - 2\alpha) = 2B \implies I^* = \frac{2B}{1 - 2\alpha}$$

- **Regime 1 ($0 \le \alpha < 0.5$):** Unique, positive, finite equilibrium $I^* > 0$. (At $B=100$, $\alpha=0.25$, $I^* = 400$).
- **Regime 2 ($\alpha = 0.5$ — The Simultaneous Singularity):** $0 \cdot I = 200$. **No finite solution exists** (`INCOHERENT_TARGET / ALGEBRAIC_DIVERGENCE`).
- **Regime 3 ($\alpha > 0.5$ — Inadmissible Negative Fixed Point):** The algebraic fixed point $I^* = \frac{2B}{1 - 2\alpha} < 0$ is strictly negative (e.g., at $\alpha = 0.75$, $I^* = -400$). Under a non-negative allocation constraint ($I \ge 0$), no admissible equilibrium exists.

B. Model 2 — Time-Lagged Discrete Dynamical System (`QC-D` / Lagged `QC-R` Across Steps $t = 0, 1, \dots, T$)

Now suppose the QC valuation oracle observes total spending G_t at period t, where $G_t = B + \alpha I_{t-1}$ (with initial seed $I_0 = 2B$ at $t=0$ when $I_{-1} = 0$), and sets $I_t = 2G_t$:

$$I_t = 2B + 2\alpha I_{t-1}$$

Unrolling this first-order linear non-homogeneous recurrence relation yields the exact closed-form trajectory for any finite horizon $t \ge 0$:

$$I_t = 2B \sum_{k=0}^{t} (2\alpha)^k = \begin{cases} \frac{2B}{1 - 2\alpha} \left(1 - (2\alpha)^{t+1}\right) & \text{if } \alpha \neq 0.5 \\ 2B(t + 1) & \text{if } \alpha = 0.5 \end{cases}$$

Furthermore, the perturbation error $e_t = I_t - I^*$ around the fixed point satisfies $e_{t} = (2\alpha)^t e_0$. This establishes four distinct dynamical regimes (verified numerically in **Table 4 of Part IV**):

1. **Monotonic Asymptotic Convergence ($0 \le \alpha < 0.5$):** Because $|2\alpha| < 1$, $(2\alpha)^{t+1} \to 0$ as $t \to \infty$, and I_t converges smoothly to $I^* = \frac{2B}{1 - 2\alpha}$.
2. **Linear Arithmetic Growth Without Division-by-Zero ($\alpha = 0.5$):** In a lagged state machine, $\alpha = 0.5$ **does not crash at $t=0$**. Instead, $I_t = 2B(t + 1)$ grows linearly by $+2B$ per year. Over a bounded 150-year horizon ($t = 0..149$ with $B=100$), every annual payment is finite ($I_{149} = 30{,}000$) and the 150-year cumulative sum is exactly $\sum_{t=0}^{149} I_t = 200 \times \frac{150 \times 151}{2} = 2{,}265{,}000\text{ test-units}$.
3. **Geometric Runaway Divergence ($\alpha > 0.5$):** Because $2\alpha > 1$, $I_t > 0$ remains strictly positive at every step (never hitting the negative algebraic fixed point $I^* < 0$) but explodes exponentially as $(2\alpha)^{t+1}$.
4. **Negative / Counter-Cyclical Dampening vs. Neutral Oscillation ($\alpha < 0$):** If QC investment displaces or crowds out baseline spending ($\alpha < 0$), convergence holds for $-0.5 < \alpha < 0$ via damped alternating oscillations. At **$\alpha = -0.5$** (Claude's boundary case), $2\alpha = -1$: the fixed point $I^* = B = 100$ exists algebraically, but the recurrence $I_t = 200 - I_{t-1}$ oscillates perpetually between $200, 0, 200, 0, \dots$ and never settles!

C. Reproducible Python Simulation & Assertion Script (`M04-feedback-sim.py`)

To satisfy Perplexity's Level `E2` (Reproducible Executable Model) standard, the following self-contained Python script verifies all five regimes and outputs the exact values recorded in Table 4:

```
def simulate_m04_feedback(B: float, alpha: float, steps: int = 150) -> dict:
    """E2 Executable Reference Model for OAL Currency Laboratory Case M04."""
    fixed_point = (2.0 * B) / (1.0 - 2.0 * alpha) if alpha != 0.5 else None
    trajectory = []
    I_prev = 0.0
    for t in range(steps):
        G_t = B + alpha * I_prev
        I_t = 2.0 * G_t
        # Closed-form verification check
        expected = 2.0 * B * (t + 1) if alpha == 0.5 else (2.0 * B * (1.0 - (2.0 *
alpha)**(t + 1))) / (1.0 - 2.0 * alpha)
        assert abs(I_t - expected) < 1e-6, f"Recurrence mismatch at t={t}"
        trajectory.append((t, round(G_t, 4), round(I_t, 4)))
        I_prev = I_t
    return {"B": B, "alpha": alpha, "fixed_point": fixed_point, "I_0":
trajectory[0][2],
            "I_1": trajectory[1][2], "I_10": trajectory[10][2], "I_149":
trajectory[-1][2],
            "cumulative_150yr": round(sum(x[2] for x in trajectory), 2)}
```

PART IV — REFERENCE TABLES & CROSS-REFERENCE APPENDIX

Table 1: Verified Six-Member File Attribution & Provenance Manifest (`P2.1`)

AI Contributor	Verified Repository Filename	Format & Length	Attribution Basis & Key Distinct Contents
ChatGPT *(OpenAI)*	`02-laboratory-charter/OAL_Trinity_Currency_Laboratory_Foundation_Package_v0.1.pdf`	PDF (21 pages, 3 Nov 2026)	Confirmed via ChatGPT thread self-identification. Contains Parts A/B/C (`F-01..F-15`, `M-01`), premise denial vs. inflation rule, and 9-stage roadmap.
Claude *(Anthropic)*	`02-laboratory-charter/OAL_Currency_Laboratory_Charter_v0.1.docx`	DOCX (23 sections, 2 Nov 2026)	Explicit header (`OAL-CURR-LAB-CHARTER-v0.1-CLAUDE-20261102`). Contains Early-Exit flowchart, Return Rules `R1–R4`, Architecture C, and Claude's 10-Case Boundary Pass.
Google Gemini *(Google)*	`02-laboratory-charter/OAL_Currency_Laboratory_Charter_and_Experimental_Baseline.pdf`	PDF / HTML (10 pages, 2 Nov 2026)	Explicit header (`Contributor: Google Gemini`). Contains 5 orthogonal matrices, QRND return schema, and Gemini's 10-Case Boundary Pass. (Amended via Section 1.4 Errata herein).
Microsoft Copilot *(Microsoft)*	`02-laboratory-charter/OAL_Currency_Laboratory_Foundational_Package_v0.1.pdf`	PDF (18 pages, 2 Nov 2026)	Confirmed via Copilot thread self-identification. Contains 5-part package, `QC-F01..06`, `BN-01..06`, `REDUNDANT`/`DIVERGENT` verdicts, and 9-folder GitHub tree.

AI Contributor	Verified Repository Filename	Format & Length	Attribution Basis & Key Distinct Contents
Perplexity AI *(Perplexity)*	`01-constitutional-baseline/OAL_Currency_Experimental_Baseline_v0_2_Perplexity.pdf` & `02-laboratory-charter/OAL_Currency_Laboratory_Charter_v0_1_Perplexity.docx`	PDF (7 pp.) & DOCX (14 sec., 2 Nov 2026)	Explicit header (`Prepared by Perplexity`). Contains `EB-01..10`, `D-01..05`, SHA-256 register, comparators `P0–P4`, cases `F01–F16`, `B01–B10`, `M01–M04`, and trace `W01`.
Grok *(xAI)*	`02-laboratory-charter/OAL_Currency_Review_Laboratory_Charter_v0.1.docx`	DOCX / PDF (20 sections, 2 Nov 2026)	Confirmed via Grok thread self-identification (also archived in `OAL_Currency_Laboratory_Conversation_Archive.docx` with Grok's 10-Case Boundary Pass).

Table 2: Canonical & Contributor-Namespaced Test ID Registry (`OAL-CURR-REG-v0.1`)

Lab Alias	Discriminating Function	Perplexity ID	Claude ID	Gemini ID	Copilot ID	ChatGPT ID
`LAB-F01`	Resource-Box Capability Discriminator (`P0` Eval, `P1` Transfer, `P2` Create, `P3` Enact)	`perplexity/v0.1/F01`	`claude/v0.1/FM-1`	`gemini/v0.1/FM-01`	`copilot/v0.1/QC-F01..04`	`chatgpt/v0.1/F-01`
`LAB-F02`	Target Resolution Locus & Temporal Value Shift (`t0` vs. `t1`, Scope Mutation)	`perplexity/v0.1/F02,F14`	`claude/v0.1/FM-2`	`gemini/v0.1/FM-02`	`copilot/v0.1/QC-F05`	`chatgpt/v0.1/F-02`
`LAB-F03`	Undefined Referent vs. Logical Contradiction Return (`UNDEFINED` vs. `INCOHERENT`)	`perplexity/v0.1/F03`	`claude/v0.1/FM-3a`	`gemini/v0.1/FM-03a`	`copilot/v0.1/BN-01,02`	`chatgpt/v0.1/F-03`
`LAB-F04`	No-Seller vs. Unknown-Seller & Non-Transferable Return (`R1–R4`)	`perplexity/v0.1/F04`	`claude/v0.1/FM-3b`	`gemini/v0.1/FM-03b,c`	`copilot/v0.1/BN-03,04`	`chatgpt/v0.1/F-04,05`
`LAB-F05`	Authorization Sensitivity (`AUTH-1` Permission-Bound vs. `AUTH-3` Sovereign Override)	`perplexity/v0.1/F05`	`claude/v0.1/FM-3c`	`gemini/v0.1/FM-01b`	`copilot/v0.1/BN-05`	`chatgpt/v0.1/F-06`
`LAB-F06`	1COIN Ablation, Premature Receipt (`W01-a`), Replay & Partial Delivery	`perplexity/v0.1/F07,F08`	`claude/v0.1/FM-4,FM-8`	`gemini/v0.1/FM-04`	`copilot/v0.1/Sec15`	`chatgpt/v0.1/F-07,15`
`LAB-F07`	NCOIN Trigger Isolation (Zero-Cost Rejection vs. Metered Cost vs. Harm)	`perplexity/v0.1/F09,F10`	`claude/v0.1/FM-5`	`gemini/v0.1/FM-05*`	`copilot/v0.1/Sec16`	`chatgpt/v0.1/F-08`

Lab Alias	Discriminating Function	Perplexity ID	Claude ID	Gemini ID	Copilot ID	ChatGPT ID
LAB-F08	Conservation vs. Net-New Value & Restorative Lifecycle (`NCOIN-X` / `NCOIN-R`)	perplexity/v0.1/F15	claude/v0.1/FM-6	gemini/v0.1/FM-06	copilot/v0.1/NCOIN-X	chatgpt/v0.1/F-09
LAB-F09	Recursive Self-Reference & Evaluator Acquisition (8-Step Budget / Fixed Point)	perplexity/v0.1/F11	claude/v0.1/FM-7	gemini/v0.1/FM-07	copilot/v0.1/QC-F06	chatgpt/v0.1/F-10
LAB-F10	Nested Simulation Referents (`Sim S` vs. `Hardware H`), Concurrency & Issuer Loss	perplexity/v0.1/F12,13,16	claude/v0.1/Phase7	gemini/v0.1/ENV-3	copilot/v0.1/Sec18	chatgpt/v0.1/F-11
LAB-M01	150-Year Macro Suite & Endogenous Feedback (`S-once`, `S-stream`, `QC-D/C/R`)	perplexity/v0.1/M01..04	claude/v0.1/Sec19	gemini/v0.1/FM-08	copilot/v0.1/Sec17	chatgpt/v0.1/F-13,14,M-01

Table 3: Summary Matrix of Gemini First-Pass Pilot Traces (`OAL-PILOT-001`)

Run / Case ID	Configuration	Pre / Post Balances (A, B, M)	Value, Execution & Settlement States	Barrier / 1COIN / NCOIN	Verdict A (Consistency) & Verdict B (Outcome)
F01-Base	P0 (Eval-Only)	$A{:}10 \backslash to\ 10$ $B{:}0 \backslash to\ 0$ (M=10)	VAL: 3 • EXEC: NOT_ATTEMPTED • SETTLE: N/A	UNSUPPORTED_OP 1C: 0 • NC: 0	**A:** PASS **B:** UNSATISFIED
F01-Base	P1 (Transfer)	$A{:}10 \backslash to\ 7$ $B{:}0 \backslash to\ 3$ (M=10)	VAL: 3 • EXEC: COMPLETED • SETTLE: FINAL	NONE 1C: 1 • NC: 0	**A:** PASS **B:** SATISFIED
F01-Base	P2 (Create)	$A{:}10 \backslash to\ 10$ $B{:}0 \backslash to\ 3$ (M=13)	VAL: 3 • EXEC: COMPLETED • SETTLE: FINAL	NONE 1C: 1 • NC: 0 or 3 (N-X)	**A:** PASS **B:** SATISFIED
F01-Base	P3 (Enact)	$B{:}0 \backslash to\ 3$ (A, M ambiguous)	VAL: 3 • EXEC: COMPLETED • SETTLE: FINAL	Collapses to P1/P2 or ID clash	**A:** INDETERMINATE **B:** SATISFIED (E0)
F01-a *(A=2, req 3)*	P1 (Transfer) vs. P2 (Create)	P1 : $A{:}2, B{:}0, M{:}2$ P2 : $A{:}2, B{:}3, M{:}5$	P1 : EXEC: BLOCKED P2 : EXEC: COMPLETED	P1 : INSUFF_INV P2 : NONE	**Discriminates P1 vs. P2**

Run / Case ID	Configuration	Pre / Post Balances (A, B, M)	Value, Execution & Settlement States	Barrier / 1COIN / NCOIN	Verdict A (Consistency) & Verdict B (Outcome)
F01-b *(Auth Denied)*	AUTH-1 vs. AUTH-3 (Sov)	AUTH-1 : $A:10, B:0$ AUTH-3 : $A:7, B:3$	AUTH-1 : BLOCKED AUTH-3 : OVERRIDE_EXEC	AUTH-1 : UNAUTHORIZED (0 NC) AUTH-3 : N-C risk	**A:** PASS **B:** BLOCKED / OVERRIDDEN
W01-a *(Transit Fault)*	Arch A (No 1C) vs. Arch B (1C-R)	$A:10 \to 7$ $B:0 \to 0$ (3 lost)	EXEC: FAILED • SETTLE: DISPUTED	1C blocked by $B \neq 3$ predicate; N-L: 3	**Proves 1COIN Tier S2 value**
F03-a *(Undefined T)*	All (`R1` Early Exit)	Unchanged	EVAL: NOT_ATTEMPTED • VAL: NONE	UNDEFINED_TARGET 1C: 0 • NC: 0	**A:** PASS **B:** BLOCKED (R1)
F03-b *($B=3 \land B=5$)*	All (`R1` Early Exit)	Unchanged	EVAL: INCOHERENT • VAL: NONE	INCOHERENT_TARGET 1C: 0 • NC: 0	**A:** PASS **B:** BLOCKED (R1)
F04-a *(Star Deed)*	All (`R2` Symbolic Val)	Unchanged	EVAL: DEFINED • VAL: V_stellar • EXEC: BLOCKED	NO_SELLER 1C: 0 • NC: 0	**A:** PASS **B:** BLOCKED (R2)
F04-b *(Unindexed Key)*	All (`R2` Val + Escrow)	Unchanged	EVAL: DEFINED • VAL: 500 • AUTH: UNKNOWN	SELLER_UNKNOWN 1C: 0 • NC: 0	**A:** INDETERMINATE **B:** PENDING_DATA

Table 4: `M04` Endogenous Feedback — Simultaneous vs. Time-Lagged Numerical Regimes ($B = 100$)

Feedback α	Simultaneous Fixed Point $I^* = \frac{2B}{1-2\alpha}$	Lagged I_0 ($t=0$)	Lagged I_1 ($t=1$)	Lagged I_{10} ($t=10$)	Lagged I_{149} (Year 150)	150-Year Cumulative Sum & Regime
$\alpha = -0.50$	*100.0 (Admissible)*	*200.0*	*0.0*	*200.0*	*0.0*	***15,000.0*** *Undamped Oscillation ($200 \leftrightarrow 0$)*
$\alpha = -0.25$	*133.33 (Admissible)*	*200.0*	*100.0*	*133.37*	*133.33*	***20,044.44*** *Damped Alternating Convergence*

Feedback α	Simultaneous Fixed Point $I^* = \frac{2B}{1-2\alpha}$	Lagged I_0 ($t=0$)	Lagged I_1 ($t=1$)	Lagged I_{10} ($t=10$)	Lagged I_{149} (Year 150)	150-Year Cumulative Sum & Regime
$\alpha = 0.00$	*200.0 (Exogenous)*	*200.0*	*200.0*	*200.0*	*200.0*	**30,000.0** *Static Constant Stream (`M01`)*
$\alpha = 0.25$	*400.0 (Admissible)*	*200.0*	*300.0*	*399.80*	*400.0*	**59,600.0** *Monotonic Asymptotic Convergence*
$\alpha = 0.50$	**Undefined ($\div 0$)**	*200.0*	*400.0*	*2,200.0*	*30,000.0*	**2,265,000.0** *Linear Arithmetic Growth (+200/\text{yr})*
$\alpha = 0.75$	*-400.0 (Inadmissible)*	*200.0*	*500.0*	*34,199.32*	*1.11 \times 10^{29}*	**3.33×10^{29}** *Geometric Runaway Divergence*